A - Many Sets

分值：400 分

Problem Statement

You are given N sequences of positive integers, each of length M. The i-th sequence is $A_i = (A_{i,1}, A_{i,2}, \dots, A_{i,M})$.

给定 N 个由正整数构成的序列，每个序列的长度均为 M。第 i 个序列是 $A_i = (A_{i,1}, A_{i,2}, \dots, A_{i,M})$。

There are M^N ways to choose one element from each of these N sequences. Find the sum, modulo 998244353, of "the number of distinct integers among the chosen elements" over all such ways.

从这 N 个序列中各选取一个元素，共有 M^N 种选取方式。对所有这些选取方式，求「所选元素中不同整数的个数」的总和，结果对 998244353 取模。

Constraints

- $1 \le N, M \le 500$
- $1 \le A_{i,j} \le NM$
- All input values are integers.
- $1 \le N, M \le 500$
- $1 \le A_{i,j} \le NM$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N M
$A_{1,1}$ $A_{1,2}$ $\dots$ $A_{1,M}$
$A_{2,1}$ $A_{2,2}$ $\dots$ $A_{2,M}$

$\vdots$

$A_{N,1}\ A_{N,2}\ \ldots\ A_{N,M}$

Output

Output the answer.

输出答案。

Sample Input 1

```
2 2
1 3
2 3
```

Sample Output 1

```
7
```

For example, if $A_{1,1}$ and $A_{2,1}$ are chosen, there are two distinct integers among the chosen elements: 1 and 2.

例如，若选取 $A_{1,1}$ 和 $A_{2,1}$，则所选元素包含两个不同整数：1 和 2。

The number of distinct integers is 1 only when $A_{1,2}$ and $A_{2,2}$ are chosen, and it is 2 in the other three cases, so the answer is 7.

只有当选出 $A_{1,2}$ 和 $A_{2,2}$ 时，不同整数的个数是 1，其余三种情况均为 2，因此答案是 7。

Sample Input 2

```
2 2
1 1
1 2
```

Sample Output 2

```
6
```

Sample Input 3

```
3 5
3 1 3 4 2
5 2 1 2 3
4 6 2 5 6
```

Sample Output 3

```
327
```